Birzeit University

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ENCS3340: ARTIFICIAL INTELLIGENCE

Project 2 Report

Comparative Study of Image Classification Using Decision Tree, Naive Bayes, and Feedforward Neural Networks

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Abstract

This project presents a comparative study of three machine learning models Naive Bayes, Decision Tree, and Feedforward Neural Network (MLP classifier) for the task of image classification. Using a curated dataset of over 500 labeled images across at least three distinct classes, each image was preprocessed by resizing (e.g., 32×32) and flattening into one-dimensional vectors. The Naive Bayes classifier, known for its simplicity and speed in high-dimensional spaces, was used as a baseline. The Decision Tree model provided interpretable classification rules, while the MLP classifier explored deeper data representations. All models were evaluated using standard metrics including accuracy, precision, recall, F1-score, and confusion matrices. The results were analyzed to assess each model's performance, generalization ability, and practical trade-offs. This study offers insights into the strengths and limitations of classic versus neural network-based classifiers in image recognition tasks.

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Introduction and dataset description

1. Introduction

Image classification is a foundational task in computer vision with applications ranging from object recognition to medical diagnostics. In recent years, machine learning techniques have significantly advanced the accuracy and efficiency of image classification systems. This project investigates the comparative performance of three distinct machine learning model, Naive Bayes, Decision Tree, and Feedforward Neural Network (MLP classifier) on a medical imaging dataset.

The primary objective is to evaluate how each algorithm handles high-dimensional visual data and to determine which method provides the most accurate and generalizable classification. Through preprocessing, model training, and thorough performance evaluation using metrics such as accuracy, precision, recall, and F1-score, the project offers insights into the practical strengths and weaknesses of each approach when applied to medical image data.

2. Dataset Description

The dataset used in this project consists of brain MRI images categorized into four classes:

- **Glioma** – a type of tumor that originates in the glial cells of the brain.
- **Meningioma** – a tumor that arises from the meninges, the protective membranes surrounding the brain and spinal cord.
- **Pituitary** – tumors located in the pituitary gland, which regulates vital hormones.
- **No Tumor** – images that show healthy brain scans without any visible tumors.

The dataset contains a total of at least 800 images for each class. All images were resized to a fixed dimension (32×32 pixels) to ensure uniformity in input size. The resized images were then flattened into one-dimensional vectors to serve as input features for the classification models. This preprocessing step allows conventional classifiers like Naive

Bayes and Decision Trees, which do not inherently handle multi-dimensional input, to effectively process the data.

The Data was stored in three files:

■ **X_flattened.npy**: This file stores the input features each image from the dataset has been resized to 32×32 pixels, converted to RGB, and then flattened into a one-dimensional array of 3072-pixel intensity values. This NumPy array contains all the processed image data in a format suitable for feeding into machine learning models.

■ **y_labels_encoded.npy**: This file contains the class labels for each image, for example, if the classes are ['glioma', 'meningioma', 'notumor', 'pituitary'], they are encoded as [0, 1, 2, 3]. Each entry in this array corresponds to the label of an image in the same order as in X_flattened.npy.

■ **label_classes.npy**: This file stores the mapping between the encoded integer labels and their original string class names. It is useful for interpreting the model's predictions and converting the output back into human-readable class names during evaluation or visualization.

Detailed explanation of each model

Naive Bayes Classifier:

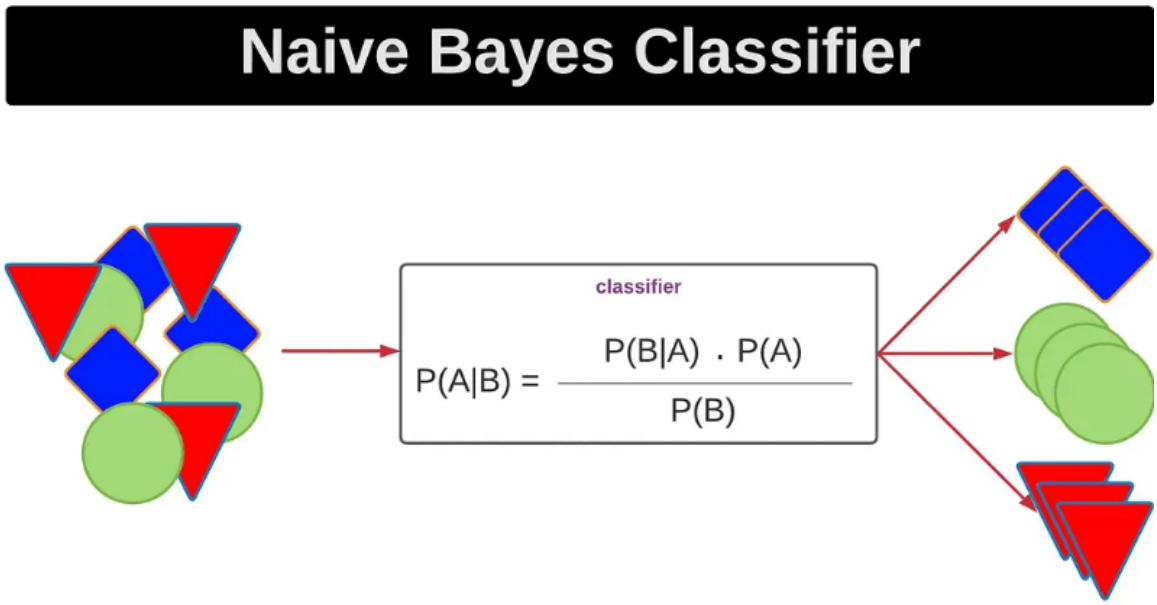

Figure 1:Naive Bayes Classifier

The Naive Bayes classifier is a probabilistic model based on Bayes' Theorem. It assumes that all features are conditionally independent given the class label. In the context of image classification, each pixel value is treated as an individual feature.

- **How it works**: For each class, the model learns the distribution of pixel values. Given a new image, it calculates the probability that the image belongs to each class and selects the one with the highest probability.
- **Pros**: Very fast to train and predict; effective with small datasets; simple to implement.
- **Cons**: Assumes pixel independence, which may reduce performance in images where pixel relationships are important.

Decision Tree Classifier:

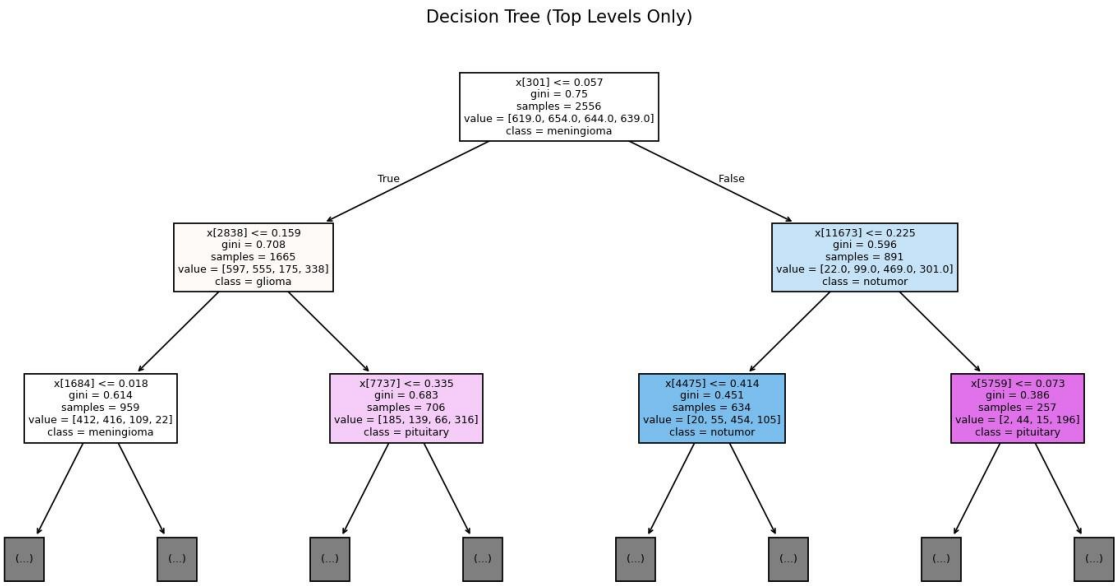

Figure 2:Decision Tree Classifier

A Decision Tree classifies data by learning a series of if-then-else decision rules based on feature values. It splits the input space into regions that correspond to specific class labels.

- **How it works**: The model recursively chooses the best feature (pixel) to split on, aiming to maximize information gain. The result is a tree where each path from root to leaf represents a decision path.
- **Pros**: Easy to understand and visualize; can capture non-linear patterns; doesn't require feature scaling.
- **Cons**: Prone to overfitting, especially with high-dimensional data like images. Requires pruning or setting a maximum depth.

Feedforward Neural Network:

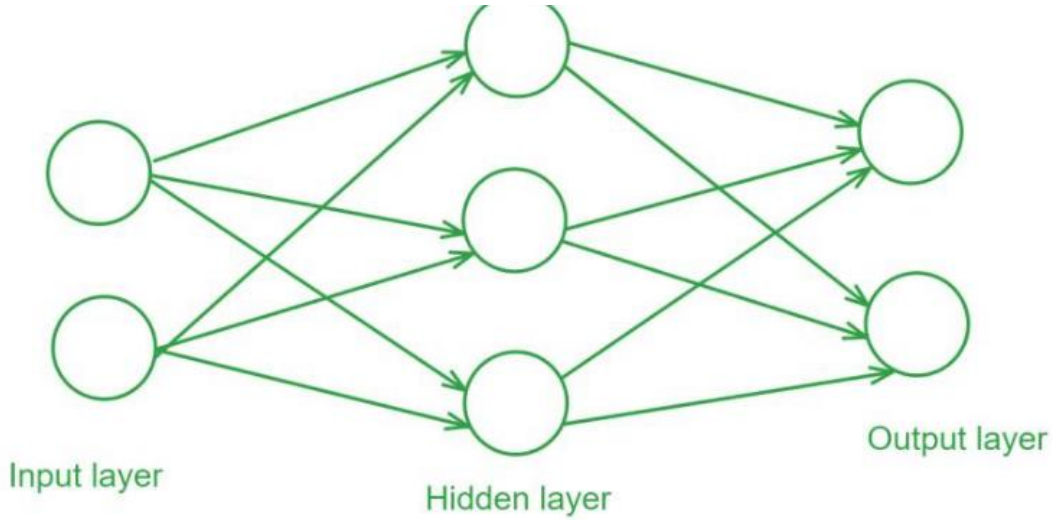

Figure 3:Feedforward Neural Network

A Feedforward Neural Network, or Multi-Layer Perceptron (MLP), is a type of artificial neural network designed to model complex relationships through layers of neurons.

- **How it works**: The input (flattened image) is passed through one or more hidden layers. Each layer applies a linear transformation followed by a non-linear activation function (e.g., ReLU). The network is trained using backpropagation to minimize prediction error.
- **Model used**: Scikit-learn's MLP classifier, typically with 1–2 hidden layers.
- **Pros**: Capable of learning complex and non-linear patterns; often achieves higher accuracy than simpler models.
- **Cons**: Requires more data and tuning; training may be slow; less interpretable than other models.

Evaluation results

Naive Bayes:

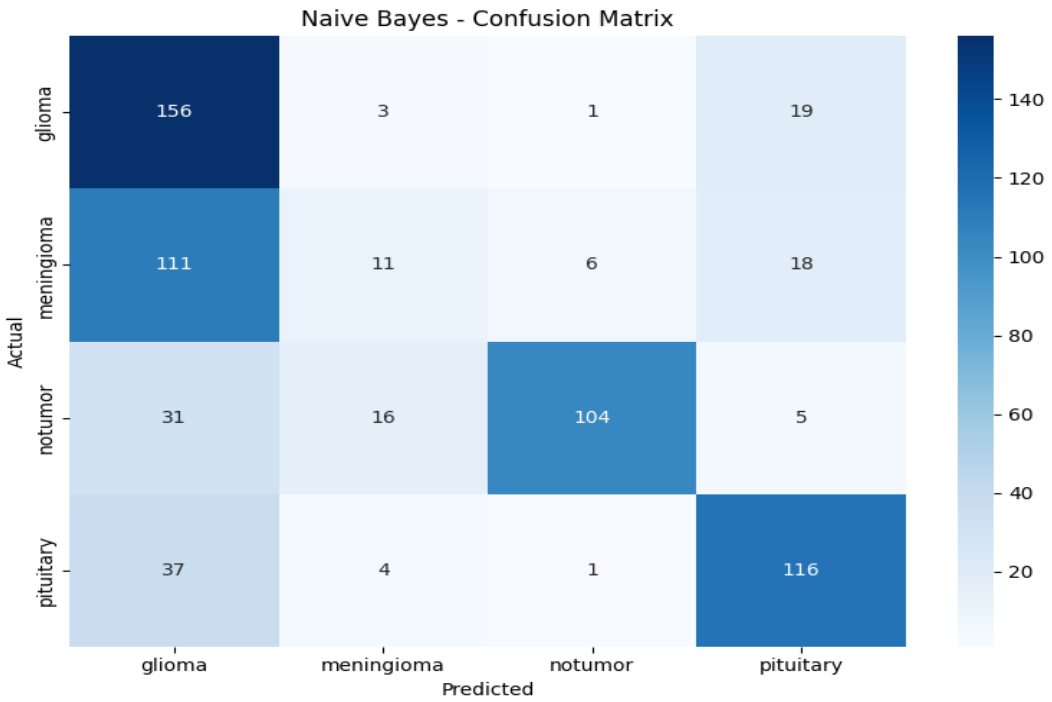

Figure 4:Confusion Matrix for Naive Bayes

Class	Precision	Recall	F1-Score	Support (How many instances)
glioma	0.47	0.87	0.61	179
meningioma	0.32	0.08	0.12	146
notumor	0.93	0.67	0.78	156
pituitary	0.73	0.73	0.73	158
Accuracy			**0.61**	**639**
Macro avg	0.61	0.59	0.56	639
Weighted avg	0.61	0.61	0.57	639

Table 1: Evaluation Results for Naive Bayes

The Naive Bayes classifier reached an overall accuracy of **61%**, which shows it's moderately effective for this image classification task. From the confusion matrix, it's clear that the model performs well on some classes but poorly on others, especially where there's visual similarity.

- **Glioma**: The model detects glioma well, with **high recall (0.87)**. This means most actual glioma cases are correctly identified. However, precision is low (0.47), so it's also labeling other tumors as glioma too often.
- **Meningioma**: This class performed the worst, with a **recall at 0.08** and a **precision of 0.32,** the model almost always misses meningioma cases and mislabels them as glioma.
- **No tumor** has **very high precision (0.93)**, meaning when the model says "no tumor," it's usually right. Recall is lower (0.67), so some no-tumor images are missed.
- **Pituitary** results are balanced, with both **precision and recall at 0.73**, indicating consistent predictions.

The macro average F1-score is 0.56, showing overall weak balance between classes.

Decision Tree:

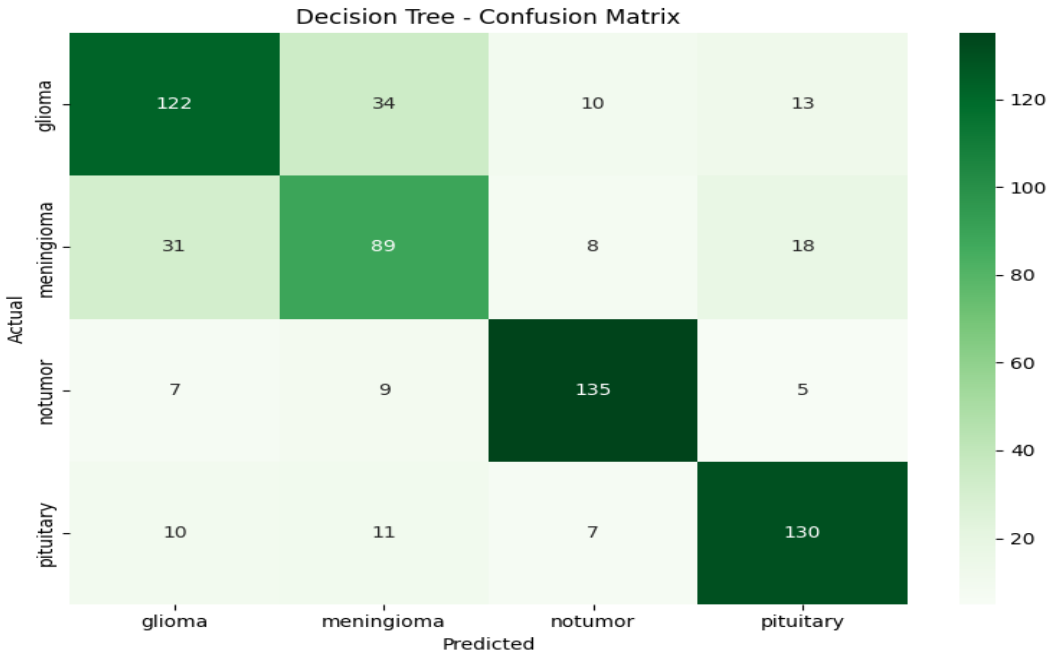

Figure 5:Confusion Matrix for Decision Tree

Class	Precision	Recall	F1-Score	Support (How many instances)
glioma	0.72	0.68	0.70	179
meningioma	0.62	0.61	0.62	146
notumor	0.84	0.87	0.85	156
pituitary	0.78	0.82	0.80	158
Accuracy			0.74	639
Macro avg	0.74	0.74	0.74	639
Weighted avg	0.74	0.74	0.74	639

Table 2:Evaluation Results for Decision Tree

The Decision Tree classifier achieved an overall accuracy of **74%**, showing a noticeable improvement over Naive Bayes and a stronger ability to handle the image classification task. The confusion matrix reflects more accurate and balanced predictions across all classes, with fewer severe misclassifications.

- **Glioma** is predicted with good consistency, achieving a **recall of 0.68**. This means the model correctly identifies most glioma cases. The **precision is 0.72**, so when it predicts glioma, it's usually right.
- **Meningioma** shows much better performance here, with both **precision (0.62)** and **recall (0.61)** being significantly higher than before. The model is now able to correctly classify the majority of meningioma cases.
- **No tumor** continues to be one of the best-performing classes, with a **precision of 0.84** and **recall of 0.87**. This means the model rarely mislabels non-tumor images, and it detects them very reliably, making this class highly accurate.
- **Pituitary** predictions are also strong and balanced, with a **precision of 0.78** and **recall of 0.82**. The classifier handles this class well, similar to the no-tumor category.

The macro average F1-score is **0.74**, which shows the model performs evenly across all classes. This improvement over the Naive Bayes macro F1-score of 0.56 indicates that the Decision Tree is better at generalizing and distinguishing between different tumor types, especially the ones that were previously misclassified.

Feedforward Neural Network:

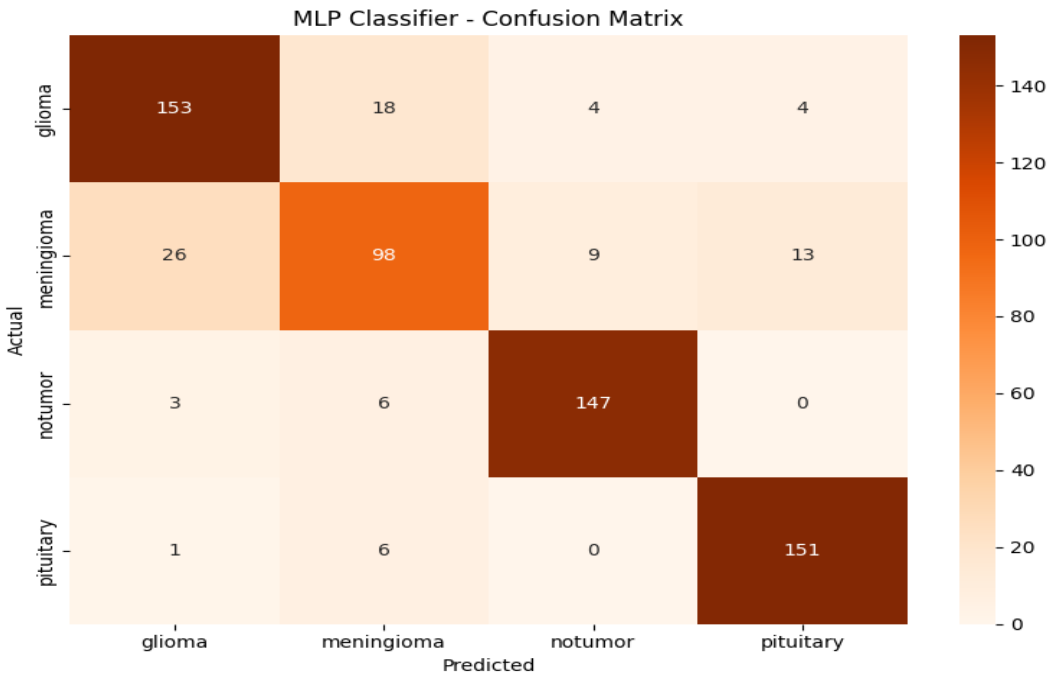

Figure 6:Confusion Matrix for Feedforward Neural Network

Class	Precision	Recall	F1-Score	Support (How many instances)
glioma	0.84	0.85	0.85	179
meningioma	0.77	0.67	0.72	146
notumor	0.92	0.94	0.93	156
pituitary	0.90	0.96	0.93	158
Accuracy			**0.86**	**639**
Macro avg	0.85	0.86	0.85	639
Weighted avg	0.86	0.86	0.86	639

Table 3:Evaluation Results for Feedforward Neural Network

The MLP classifier achieved an **overall accuracy of 86%**, outperforming the Decision Tree and Naive Bayes models. This highlights its superior capability in learning complex patterns in medical image data. The confusion matrix shows strong and well-distributed predictions across all tumor types, with minimal critical misclassifications.

- **Glioma** was predicted with high reliability. It achieved a **precision of 0.84** and a **recall of 0.85**, indicating that the classifier is both consistent in detecting glioma cases and generally correct when it predicts them.
- **Meningioma** remains a challenging class but sees reasonable performance with a **precision of 0.77** and **recall of 0.67**. While some meningioma cases are still confused with glioma and pituitary, the MLP does a better job compared to earlier models, with improved handling of false negatives.
- **No tumor** is one of the best-performing categories, with a **precision of 0.92** and **recall of 0.94**.
- **Pituitary** tumor classification is also highly accurate, with a **precision of 0.90** and **recall of 0.96**, indicating very few misclassifications.

The **macro average F1-score is 0.85**, and the **weighted average is also 0.86**, showing that the MLP classifier is not only accurate but also fair across all classes, without heavily favoring any one category. This marks a significant improvement in generalization and robustness over simpler models like Naive Bayes, especially in complex classification tasks like brain tumor detection.

Comparative analysis of models

1. Feedforward Neural Network (MLP classifier) – Best Performance

- **Reason**: Neural networks are designed to capture complex and non-linear relationships in data. Since images have patterns, textures, and pixel dependencies, MLPs are better suited to learn these features.
- **Advantage**: Unlike simpler models, the MLP can learn from the entire structure of the image, even after it's flattened, due to its multi-layer architecture.
- **Result**: It performed best overall, especially for challenging classes, showing high accuracy and balanced precision and recall.

2. Decision Tree – Moderate Performance

- **Reason**: Decision trees split the data based on feature thresholds (e.g., pixel values), which helps detect simple decision rules.
- **Advantage**: It works well for clearly separable patterns and does not assume any distribution.
- **Limitation**: It struggles with high-dimensional input like images and can easily overfit unless carefully pruned.
- **Result**: Better than Naive Bayes but less consistent than MLP, especially with more complex or noisy image data.

3. Naive Bayes – Lowest Performance

- **Reason**: Naive Bayes assumes that all features (pixels) are independent, which is not true in image data where neighboring pixels are highly related.
- **Advantage**: Very fast and simple, good for baseline comparisons.
- **Limitation**: Cannot model the spatial or structural patterns in images effectively.
- **Result**: Underperformed because it couldn't capture the real complexity in MRI image classes.

Conclusion

This project compared three models Naive Bayes, Decision Tree, and Feedforward Neural Network for classifying brain MRI images. The Feedforward Neural Network gave the best overall results, showing strong performance on most classes due to its ability to learn complex patterns. The Decision Tree performed moderately, while Naive Bayes had the lowest accuracy due to its simplifying assumptions. In summary, neural networks are more effective for image classification tasks, especially when the data has complex or non-linear features.